Freight Rate Prediction

Validation, Modeling, and December Forecast

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Watch the Loom video walkthrough

1 Problem and input contracts

The development file contains 48,000 labeled loads from January 1 to October 31, 2025. The final file contains 12,000 November–December loads with hidden prices. The main output predicts total dollars for every template ID and contains exactly `load_id,predicted_rate`. No final labels are available for model selection.

The separate December file provides 31 dates for one fixed load. It lacks market index and coordinates, so a model evaluated on those reduced inputs is used. The supplied `score.py` validates both outputs and produces the December chart in Section 6; it does not calculate predictive accuracy.

2 Train–validation splits and evaluation

Why chronological splits? Final predictions are one to two months after training. Random splits would mix future and past regimes and understate the challenge of forecasting time effects. Expanding training windows followed by two-month evaluation windows reproduce the intended horizon. All rows on a date remain on the same side of each cutoff.

Stage	Training dates (2025)	Evaluation dates (2025)
Rolling A	Jan 1–Apr 30	May 1–Jun 30
Rolling B	Jan 1–May 31	Jun 1–Jul 31
Rolling C	Jan 1–Jun 30	Jul 1–Aug 31
Rolling D	Jan 1–Jul 31	Aug 1–Sep 30
Development confirmation	Jan 1–Aug 31	Sep 1–Oct 31
Final refit and inference	Jan 1–Oct 31	Nov 1–Dec 31

Training-only preprocessing. Medians, spline knots, scales, and category vocabularies are fitted inside each training partition. IDs and price labels are excluded from features. Final covariates are used only for descriptive inspection and inference; they do not fit preprocessing or select the recipe.

Aggregation and selection. Errors from repeated folds are averaged within each month; May–September then receive equal weight. The fixed comparison has seven candidates per input contract. Eligibility requires at least 2% macro-MAE improvement over the matched CatBoost baseline, no month more than 5% worse, and no macro-RMSE increase above 5%. The eligible winner minimizes macro MAE, then must pass a withheld-city check (Section 5).

Evaluation scope. Rolling windows overlap and are correlated. September also appears in model selection, and September–October results were reviewed during development. These are development backtests, not independent test scores. Final November–December labels are unavailable.

3 Exploratory analysis and model design

3.1 Data quality and price relationships

Distance strongly tracks total price, while observed price per mile falls with distance. This supports smooth distance effects rather than a constant rate per mile. Figure 1 includes all original loads; its band is an interquartile spread, not a confidence interval.

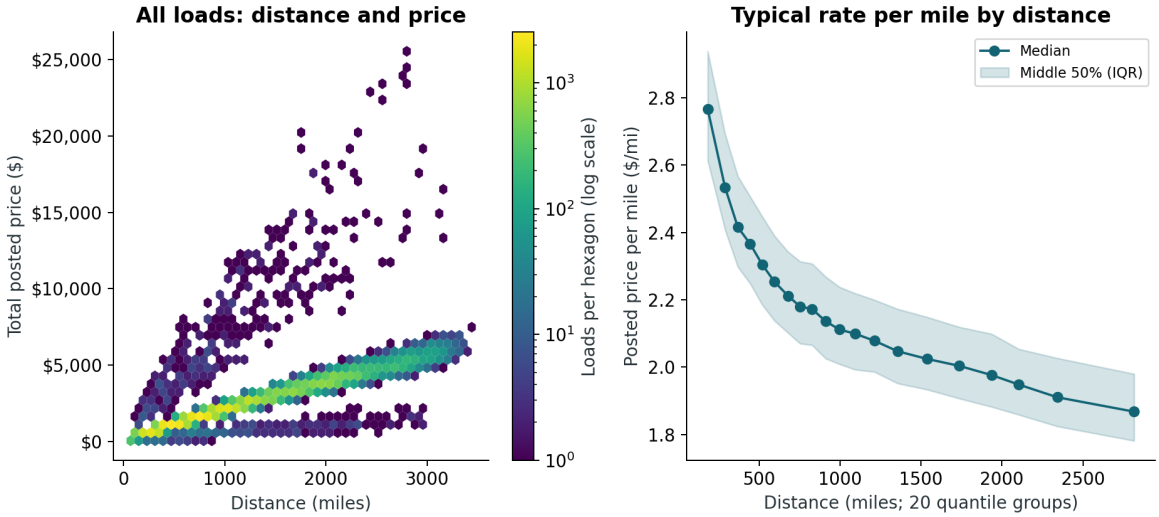


Figure 1: Distance, total price, and price per mile in the labeled development data.

Development inputs contain 300 missing and 292 negative weights, plus 374 missing market values. Final inputs contain 165 missing and 145 negative weights, plus 249 missing market values. Nonpositive weights become missing with separate original-missing and invalid-weight flags. Imputation uses training statistics. **All 48,000 original price labels are retained.**

The quote signal's association with actual price per mile reverses across months and is near zero in August. Both deployed models exclude it. Unknown provenance does not establish leakage; final usefulness cannot be measured without labels.

3.2 Main structured forecast

For a load with distance d, the model predicts log price per mile and converts back to total dollars:

$$z = \log(\text{posted_rate}/d), \qquad \widehat{\text{rate}} = d \exp\Big(\widehat{g}(x) + \widehat{h}(x)\Big).$$

The structured component g uses cubic splines for distance, valid weight, and supplied coordinates; shared endpoint-city and more strongly shrunk directed-lane effects; equipment; market index and missingness; weekly terms; an extrapolating linear trend; and quarter-position hinges with equipment interactions. Unknown city and lane encodings contribute zero in this component.

Up to eight Huber-style ridge reweighting steps reduce extreme-label influence while keeping every training weight positive. The residual component h is a 250-tree, depth-five CatBoost model fitted on in-sample log residuals. It excludes elapsed days so the structured trend remains the explicit extrapolating component.

Residual benefit is modest. Rolling MAE improves from \$86.58 to \$85.69 (about 1%). September–October reverses the ordering: \$89.43 without residual trees versus \$90.15 with them. Most improvement comes from the structured model. The frozen selection is retained; the 2% eligibility rule compares candidates with the matched baseline, not each intermediate model.

3.3 Reduced December model

The reduced model is a 600-tree, depth-six calendar CatBoost regressor with dollar MAE loss. It uses only supplied December inputs and deterministic derivatives. Reduced structured candidates fail temporal stability checks; no market values or coordinates are fabricated.

4 Results and error analysis

4.1 All-row confirmation comparison

The following scores use January–August training and the same **9,523 September–October loads**. No evaluation target is deleted, repaired, or filtered.

Model / input contract	MAE ($)	RMSE ($)	WAPE (%)	MAPE (%)
Selected main / full	**90.15**	629.98	3.77	3.97
CatBoost baseline / full	118.06	637.01	4.93	5.10
Selected December / reduced	**109.70**	633.93	4.59	4.94
CatBoost baseline / reduced	113.69	635.69	4.75	5.00

MAE is mean absolute dollar error. RMSE gives large errors greater weight by squaring before averaging. WAPE divides total absolute error by total actual dollars; MAPE averages each load's absolute percentage error. Main mean signed error is −$49.94 (prediction minus actual). The hidden scoring metric is unspecified.

4.2 Rolling results and component contributions

Main rolling macro MAE is $85.69 versus $121.23 (29.3% improvement). Reduced calendar CatBoost achieves $111.95 versus $129.16 (13.3%). Removing quarter terms or the trend raises full structured macro MAE to $117.67 or $110.33.

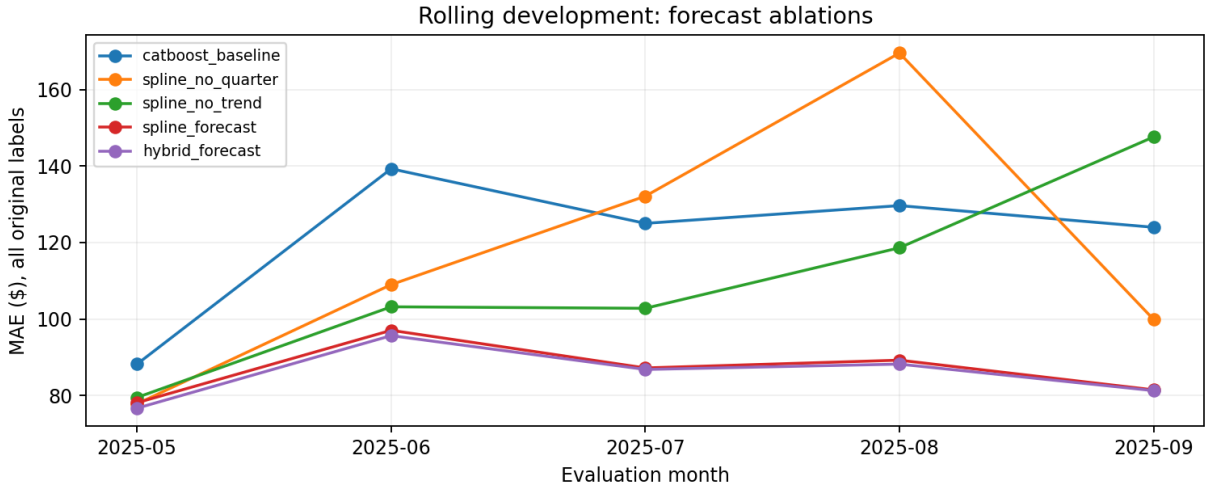


Figure 2: Rolling monthly MAE, averaging repeated folds within each month. Trend and quarter terms contribute substantially; residual trees add a small margin.

4.3 Tail diagnostic

For error analysis only, define a tail using *actual* price per mile above twice, or below half, its equipment-month median. This target-defined diagnostic identifies 142 of 9,523 loads (1.49%). It is not a preprocessing or scoring rule.

Population	Rows	MAE ($)	RMSE ($)	MAPE (%)
All original loads	9,523	90.15	629.98	3.97
Remaining diagnostic subset	9,381	34.54	70.79	1.47

The 142 tail loads contribute 62.25% of absolute error and 98.76% of squared error. RMSE remains valid, but is less responsive to gains elsewhere. The subset's 1.47% MAPE does not replace the 3.97% headline score. These observations establish neither an irreducible noise floor nor an inability to predict extremes.

5 Unfamiliar inputs and forecast limitations

5.1 City withholding

Final inputs contain eight new cities: Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, and San Diego. New endpoints affect 1,447 loads (12.1%); 1,461 loads use unseen directed lanes. Ordinary rolling windows contain no new cities, motivating a separate stress test.

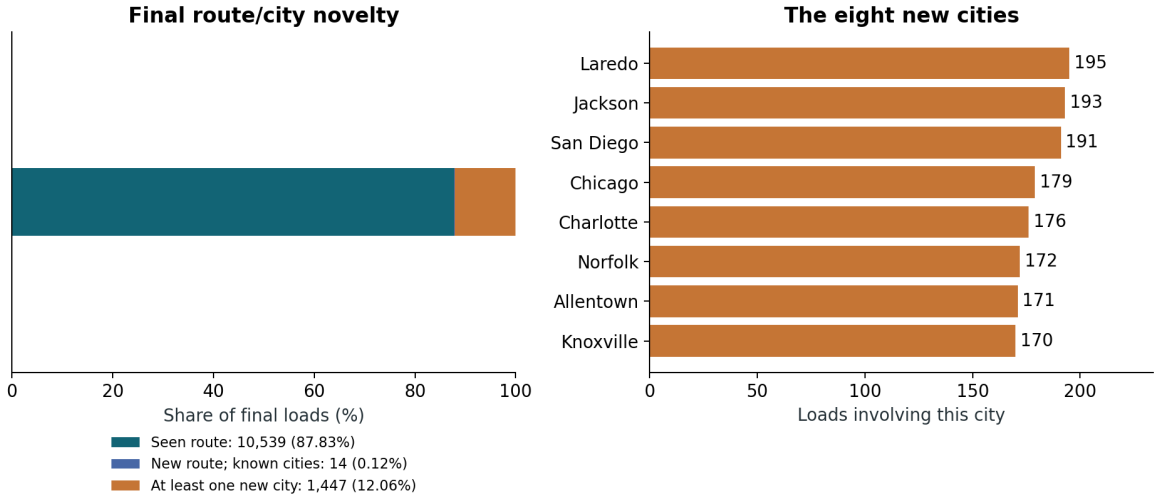


Figure 3: Endpoint and lane novelty in final inputs, and counts of loads involving each new city. These describe inputs, not predictive accuracy.

A target-independent SHA256 ordering selects Buffalo, Houston, Memphis, and St. Louis. Removing their loads from both endpoint roles leaves 25,679 January–June training rows. Evaluation uses the same 1,040 affected July–August loads in the following comparison.

Same affected evaluation population	MAE ($)
Selected main, city histories withheld	**51.35**
Matched full-input baseline, city histories withheld	102.55
Main, all January–June training cities available	48.51

The stress result is 5.9% worse than the same-row counterfactual. Withholding also changes the training population, so this does not isolate a causal city effect or establish accuracy on the eight actual new cities. Reduced-model stress MAE is $95.50 versus $140.24 for its baseline. Both selected models pass the city gate of at most 10% worse MAE than their matched baseline.

5.2 Remaining uncertainty

Forecast-horizon drift. Within the target-defined non-tail subset, September MAE is $29.19 with mean signed error −$2.78; October is $39.71 and −$30.22. Refit through October refreshes the level, but two-month drift can recur.

Quarter recurrence. Ablations support the calendar hypothesis in development data. Three completed quarters cannot establish annual seasonality, a Q4 ramp, or holiday behavior. Full-input forecasts also remain conditional on supplied market values.

Missing market values. The full model uses a training median and a missingness flag. The 73-load confirmation slice is biased upward. Two bounded training-only alternatives yield rolling macro MAE of $85.75 (recent 28-day median) and $84.85 (calendar estimate), versus current $85.69. Neither meets the predeclared 2% replacement gain; the calendar estimate also worsens the missing-market slice in three of four folds. Current imputation is retained.

6 Fixed December prediction chart

The supplied scenario fixes **Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb**. Only the date changes across December 1–31. Predictions come directly from the selected reduced calendar CatBoost model.

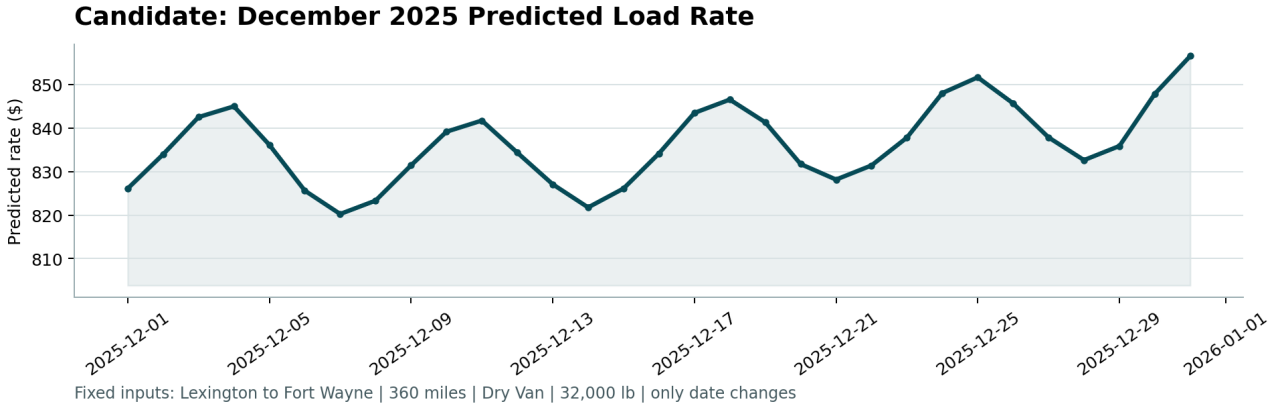


Figure 4: Fixed December prediction chart produced by the unchanged supplied `assessment/score.py`, using `outputs/december_predictions.csv`.

Prices range from **$820.25 to $856.53**. The chart shows the model's conditional date response, reflecting learned weekday and quarter-position effects. It is not measured December accuracy.

6.1 Chart provenance and output validation

The chart in Figure 4 is the supplied scorer's output, `candidate_december.png`, embedded directly. It is not a redrawn chart. The scorer is unchanged from the original assessment attachment. To reproduce it from the repository root after setup:

```
.venv/bin/python assessment/score.py \
  --predictions validation_predictions.csv \
  --december-predictions outputs/december_predictions.csv \
  --output-dir reports/assets
```

Check	Result
Main schema and IDs	Exactly two columns; all 12,000 unique template IDs in template order
Main prices	Finite and positive; $194.15–$7,073.21; no $1 floor activations
December contract	All 31 dates; original columns, order, and fixed inputs preserved
Training coverage	Both recipes refitted on all 48,000 original labeled loads
Supplied scorer	Both files accepted; required December chart generated

6.2 Reproducibility

A clean filesystem export was tested in a newly created Python 3.12.3 environment. All pinned dependencies installed, `pip check` passed, and all 26 tests passed. EDA regenerated eight figures; model refits, prediction, and scoring completed. Both regenerated prediction CSVs are **byte-for-byte identical** to the submitted files. The check used included inputs and the frozen recipe, without existing fitted models or review files; its temporary environment was removed.

The root README provides complete setup and run commands. Saved protocol, recipe, and evaluation evidence are included under `evaluation/`. Original inputs and the scorer are included once. Drafts, reviews, experiment logs, local environments, and fitted binaries remain outside the submission allowlist.